TEAM DETAILS

Team Details

TEAM NAME

The VK7

TEAM LEADER

Vikas Kumar

TEAM SIZE

1 (solo)

PROBLEM STATEMENT

Intelligent Conversational AI for KSP Crime Database

THE PROBLEM IN ONE LINE

A single repeat offender appears in the FIR database under **21 different spellings** across 17 police stations. Exact-match search returns 2 of them. The other 19 are invisible to the investigator.

BRIEF ABOUT THE SOLUTION

An investigation-grade identity layer, not another chatbot

Chat is the interface. The product is the trust machinery underneath it.

WHAT IT DOES

Resolves the **same human across FIR records** despite transliteration, script changes, name-order flips, initials, concatenation and age-entry drift.

An investigator types a name in any form — Latin or Kannada — and gets one consolidated identity with every linked record.

WHY IT MATTERS

Karnataka runs 1,100+ police stations entering names by hand. The same person becomes Ravi Patil, R. Patil, Patil Ravi, ರವಿ ಪಾಟೀಲ್.

Every missed link is a missed pattern — and a missed lead.

WHAT MAKES IT SAFE

Every match carries its **reasoning, provenance and confidence**. The system refuses to answer below threshold rather than guessing.

In policing, a fabricated link is worse than no link.

LIVE QUERY · KANNADA INPUT

2

21

OPPORTUNITIES

How this differs, and why it holds up

A · HOW DIFFERENT IS IT?

The obvious build for this challenge is **natural language → SQL → results table**. That assumes the database is already clean.

It is not. We attack the layer underneath: **who is actually the same person**. No amount of query sophistication recovers a record that name search cannot reach.

B · HOW DOES IT SOLVE THE PROBLEM?

Kannada→Latin transliteration, an **Indian-name phonetic key** (not Soundex), IDF-weighted token alignment, and average-linkage clustering that refuses to chain distinct identities through one ambiguous record.

Result: **99.2% recall** against ground truth, versus 17.9% for exact match.

C · USP

Explainability is not a feature bolted on — it is the output format.

Every response ships the tables read, the method, the thresholds, the evidence per link, and an explicit statement of what the system could not determine.

It is built to be defended in court, not just demoed on stage.

RECALL VS EXACT MATCH

5.5×

more true links recovered

SEARCH SPACE REDUCTION

92.2%

fewer comparisons than brute force — scales to state volume

WARM QUERY LATENCY

15–30ms

index built once at cold start, then cached

LIST OF FEATURES OFFERED

What the solution does today

IDENTITY RESOLUTION

- Kannada → Latin transliteration, syllable-aware (inherent vowels, virama)
- Phonetic key tuned to Indian transliteration — aspiration, retroflex/dental, vowel length
- Name-order flips, initials, concatenation, casing and whitespace damage
- Age-drift tolerance via implied birth year, with gender as a hard signal
- IDF weighting so common surnames cannot manufacture links

INVESTIGATIVE SURFACE

- Consolidated case timeline per resolved identity
- Cross-district and cross-station spread of one offender
- Offence-pattern breakdown per identity
- Side-by-side contrast with what exact search would have returned
- Per-link reasoning in plain language

TRUST & GOVERNANCE

- Provenance on every response — tables, method, thresholds, latency
- Confidence banding: high-confidence / probable / needs-review
- **Explicit refusal** below threshold instead of a plausible guess
- Stated limitations returned alongside every result

ENGINEERING

- Phonetic blocking — near-linear instead of $O(n^2)$
- Zero runtime dependencies in the resolution engine
- Benchmarked against held-out ground truth, not asserted
- Deployed end to end on Catalyst

PROCESS FLOW

Query to defensible answer

Every stage is inspectable. Nothing is a black box between the investigator and the record.

1. **1 · Input** — Any script, any spelling
2. **2 · Normalise** — Transliterate, strip noise
3. **3 · Block** — Phonetic keys, 92% pruned
4. **4 · Compare** — IDF-weighted alignment
5. **5 · Cluster** — Average linkage, no chaining
6. **6 · Explain** — Reasoning + provenance

Investigator types a name in Kannada or Latin, complete or partial. No query syntax, no schema knowledge, no dropdown filters required.

Design rule

The system must be able to say "I don't know".
A wrong link is worse than no link.

Consolidated identity + case timeline

confidence · evidence per link · tables read · stated limitations

Below threshold → explicit refusal

"No identity meets the confidence threshold." Never a plausible guess.

INTERFACE ANATOMY

The evidence ledger

The interface is modelled on a case docket, not a chat window. The argument is visible at a glance.

EXACT-MATCH SEARCH

2 records returned

RESOLVED IDENTITY

21 records · 13 districts · 17 stations

Ravi Patil

confidence 0.97

CLUSTER C0003 · 7 RECORDED SPELLINGS

REGISTERED	NAME AS RECORDED		AGE	STATION
2023-01-03	RAVI PATIL	MISSED	32	Shivamogga East PS
2023-04-07	R. Patil	MISSED	32	Raichur West PS
2024-04-08	Patil Ravi	MISSED	32	Davanagere East PS
2024-12-03	ರವಿ ಪಾಟೀಲ್		32	Bengaluru City East PS

Names render in monospace so invisible data-entry damage — trailing spaces, double spaces, casing — becomes visible. Every row exact search missed carries a red edge.

Deployed entirely on Catalyst

Designed so the model layer can be replaced without touching business logic — LLMs turn over, a police system does not.

PRESENTATION · CATALYST WEB CLIENT HOSTING

Investigator console

Evidence ledger · confidence banding · provenance panel · same origin as API, so no CORS surface

APPLICATION · CATALYST SERVERLESS — ADVANCED I/O FUNCTION (NODE.JS)

Normalisation

Kannada transliteration, token repair, vocabulary

Resolution engine

Blocking, IDF alignment, average-linkage clustering

Explanation layer

Reasoning, confidence, provenance, refusal

Semantic API

Governed contract — no raw SQL exposed

DATA · CURRENT

Bundled synthetic FIR corpus

1,400 FIRs · 1,663 accused records · 72 stations · schema-faithful to the KSP ER diagram
Index built once at cold start, cached in module scope

DATA · MIGRATION PATH

Catalyst Data Store · QuickML · Zia

Relational store for live FIR data, QuickML for natural-language and RAG, Zia speech services for voice. Engine is storage-agnostic by design.

TECHNOLOGIES USED

Stack

RUNTIME & DELIVERY

- Node.js 24 — Catalyst Advanced I/O Function
- Express — HTTP routing inside the function
- Vanilla HTML / CSS / JS — no framework, no build step
- Catalyst CLI — deployment pipeline

RESOLUTION ENGINE

- Custom Kannada→Latin transliteration (Unicode-aware)
- Custom phonetic algorithm for Indian names
- Jaro–Winkler string similarity
- IDF term weighting
- Union-Find with average-linkage constraint
- **Zero runtime dependencies** — 545 lines, fully auditable

DATA GENERATION & EVALUATION

- Python 3 + SQLite — schema-faithful synthetic FIR corpus
- Deterministic seeding — fully reproducible dataset
- Held-out ground-truth clusters for scoring
- Pairwise precision / recall / F1 harness

DELIBERATE NON-CHOICES

No vector database, no embedding model, no LLM in the resolution path.

Identity resolution must be **deterministic and reproducible** — the same query must return the same answer, and every link must be explainable to a court. A stochastic model cannot offer that guarantee.

LLMs are reserved for the conversational layer, where variance is acceptable.

CATALYST SERVICES USED

Catalyst services

SERVICE	STATUS	ROLE IN THE SOLUTION
Catalyst Serverless Advanced I/O Function	In use	Hosts the entire resolution engine and the semantic API. Node.js 24 runtime, 512 MB.
Catalyst Web Client Hosting	In use	Serves the investigator console. Same origin as the function, which removes the cross-domain surface entirely.
Catalyst CLI	In use	Project initialisation and deployment pipeline.
Catalyst Data Store	Next	Relational store for live FIR data. The engine already reads through a storage-agnostic record interface, so this is a swap, not a rewrite.
Catalyst QuickML LLM serving + RAG	Next	Natural-language question layer above the resolved identity graph, and RAG over case documents.
Catalyst Zia Services	Roadmap	Speech-to-text and translation for Kannada voice queries from the field.
Catalyst Authentication + API Gateway	Roadmap	Role-based access enforced at the data layer, with throttling in front of the function.

Deployment is entirely on Catalyst. Stated status is accurate as of submission — nothing listed as "in use" is aspirational.

ESTIMATED IMPLEMENTATION COST

Cost profile

Indicative, for a statewide rollout on Catalyst.

PROTOTYPE · TODAY

~0

Runs inside the hackathon Catalyst credit allocation. One serverless function, one hosted client, no persistent store.

PILOT · ONE DISTRICT

Low

Data Store for live FIRs, function invocations, QuickML for the language layer. Index rebuild is a scheduled job, not a per-request cost.

STATEWIDE

Moderate

Dominated by Data Store volume and LLM serving. The resolution engine itself stays cheap — blocking keeps it near-linear rather than quadratic.

THE ARCHITECTURAL COST DECISION

Identity resolution runs as a **batch index build**, not per query. Adding records re-indexes incrementally; investigator queries hit a cached structure and cost almost nothing.

The alternative — calling an LLM per query to reconcile names — would scale cost linearly with usage, and would not be reproducible. This design keeps the expensive work bounded and the frequent work free.

SNAPSHOTS OF THE PROTOTYPE

The prototype, live on Catalyst

https://ksp-crime-intel-60079832998.development.catalystserverless.in/app/index.html

RESOLVED IDENTITY — KANNADA QUERY

EXACT-MATCH SEARCH

2

records returned for "ರವಿ ಪಾಟೀಲ್". 19 further records belonging to the same person were never surfaced.

RESOLVED IDENTITY

21

records across 13 districts and 17 stations, 2023–2026.

Ravi Patil

CLUSTER C0003 · 8 RECORDED SPELLINGS

confidence 0.97 · high-confidence

REGISTERED	NAME AS RECORDED	AGE	STATION	OFFENCE
2023-06-05	Ravi Patil· MISSED	32	Vijayapura East PS	Vehicle Theft
2023-04-07	R. Patil MISSED	32	Raichur West PS	Robbery
2023-10-18	RAVI PATIL MISSED	33	Hassan West PS	Theft
2023-07-14	R. Patil MISSED	32	Hubballi-Dharwad East PS	Vehicle Theft
2023-01-03	RAVI PATIL MISSED	32	Shivamogga East PS	Vehicle Theft
2024-05-31	RAVI PATIL MISSED	33	Bengaluru Rural North PS	Illegal Arms Possession
2024-05-23	Ravi P MISSED	34	Hassan West PS	Robbery
2024-12-29	Ravi· Patil MISSED	33	Mysuru South PS	Burglary
2024-12-03	ರವಿ ಪಾಟೀಲ್	32	Bengaluru City East PS	Dacoity
2024-05-18	Ravi· Patil MISSED	32	Shivamogga North PS	Theft

DELIBERATE REFUSAL BELOW THRESHOLD

NAME AS YOU HAVE IT

Zzzq Nobody

Resolve identity

ರವಿ ಪಾಟೀಲ್ · Ravi Patil · R. Patil · Khan Yusuf · YusufKhan · a name not on file

No identity meets the confidence threshold

No identity cluster met the confidence threshold. This is a deliberate refusal, not an empty database — narrow or correct the spelling and retry.

1,400 FIRs · **1,663** accused records recall **99.2%** vs **17.9%** exact match precision **84.7%**

92.2% fewer comparisons than brute force

PROTOTYPE PERFORMANCE REPORT

Benchmarking

Measured against held-out ground truth on 1,663 accused records. Not asserted — scored.

PAIRWISE ACCURACY VS GROUND TRUTH

METHOD	PRECISION	RECALL	F1
Exact string match	34.1%	17.9%	23.5%
This engine	**84.7%**	**99.2%**	**91.3%**

Residual precision loss is **data-limited, not model-limited**: the corpus contains genuinely distinct people sharing a name and age. No method separates those without a corroborating identifier. The system surfaces them for review rather than asserting a link.

RECALL

99.2%

vs 17.9% exact match — 5.5× more true links recovered

SCALE BEHAVIOUR

108K

comparisons instead of 1.38M — 92.2% pruned by phonetic blocking

LATENCY

~2s cold · 15–30ms warm

Well inside the 30-second function ceiling

LINKS

Submission links

1 · GITHUB PUBLIC REPOSITORY

https://github.com/«YOUR-USERNAME»/ksp-identity-resolution

2 · DEMO VIDEO (3 MINUTES, PUBLIC)

«DEMO VIDEO URL»

3 · DEPLOYED LINK — CATALYST

https://ksp-crime-intel-60079832998.development.catalystserverless.in/app/index.html

Deployment is exclusively on the Catalyst platform, as required.

Built for a decade, not a demo

Four decisions that matter more than any single feature.

1 · SEMANTIC LAYER BETWEEN MODEL AND SCHEMA

The language model never writes raw SQL. It generates against a governed set of defined entities and canonical joins. When the schema changes — and over a decade it will, repeatedly — you update one layer, not every prompt.

2 · MODEL-AGNOSTIC ADAPTER

LLMs turn over every few months. Nothing in the business logic knows which model it is calling. Swapping providers is a configuration change.

3 · GRACEFUL DEGRADATION

If the language layer is unavailable or low-confidence, the system falls back to structured identity search and still works. A decade-scale police system cannot have a single AI dependency as a hard failure point.

4 · AUDIT OF THE QUERIES THEMSELVES

Police search logs are themselves sensitive. Who searched for whom, when, and under what case authority belongs in an append-only store from day one — not bolted on after an incident.

NEXT BUILD ORDER

Data Store migration → **QuickML natural-language layer over the resolved identity graph** → **criminal network graph** (co-accused links across resolved identities) → **Zia voice input in Kannada** → **role-based access enforced at the data layer**.